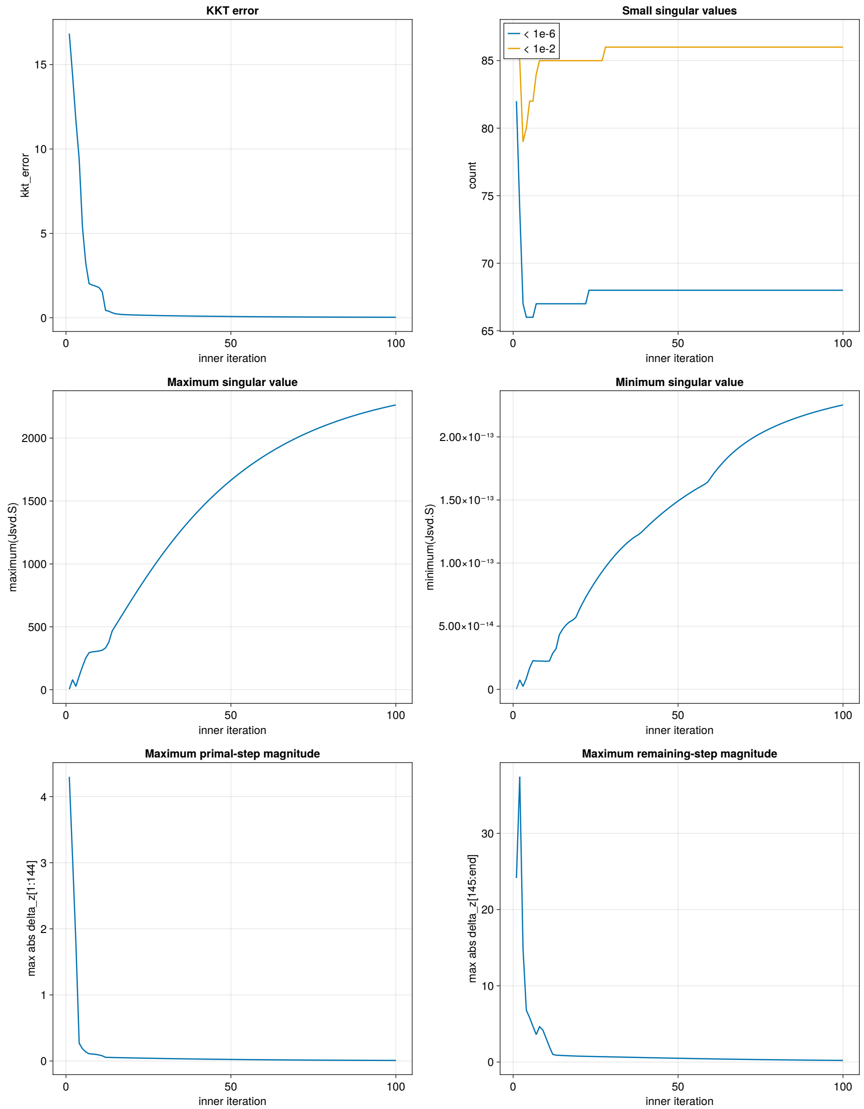
Solver Diagnostics

Recorded 100 inner iterations; reporting 100 rows (cap: 1000).

Summary statistics

Metric	Minimum	Maximum	Mean	Median
`kkt_error`	3.041494e-02	1.683624e+01	7.810390e-01	7.514200e-02
`count(Jsvd.S < 1e-6)`	6.600000e+01	8.200000e+01	6.797000e+01	6.800000e+01
`count(Jsvd.S < 1e-2)`	7.900000e+01	8.700000e+01	8.557000e+01	8.600000e+01
`maximum(Jsvd.S)`	4.591747e+00	2.263635e+03	1.466130e+03	1.675823e+03
`minimum(Jsvd.S)`	1.118702e-16	2.253136e-13	1.387105e-13	1.500900e-13
`maximum(abs.(delta_z[1:144]))`	8.288486e-03	4.300357e+00	1.241102e-01	2.335358e-02
`maximum(abs.(delta_z[145:end]))`	2.105846e-01	3.740870e+01	1.541555e+00	4.921677e-01

Metrics versus inner iteration



Per-iteration diagnostics

Iter	KKT error	$N_{<10^{-6}}$	$N_{<10^{-2}}$	$\sigma_{\max}$	$\sigma_{\min}$	$\max\lvert\delta z_{1:144}\rvert$	$\max\lvert\delta z_{145:}\rvert$
1	1.683624e+01	82	87	4.591747e+00	1.118702e-16	4.300357e+00	2.413179e+01
2	1.437564e+01	74	85	7.997262e+01	7.436759e-15	3.064873e+00	3.740870e+01
3	1.170472e+01	67	79	2.635557e+01	2.240684e-15	1.792523e+00	1.479713e+01
4	9.399197e+00	66	80	1.086595e+02	8.618646e-15	2.693541e-01	6.792953e+00
5	5.352853e+00	66	82	1.843720e+02	1.691805e-14	1.847728e-01	5.826102e+00
6	3.223240e+00	66	82	2.522011e+02	2.263253e-14	1.374174e-01	4.690842e+00
7	2.025206e+00	67	84	2.941787e+02	2.241585e-14	1.095958e-01	3.599822e+00
8	1.936269e+00	67	85	3.013408e+02	2.237905e-14	1.058514e-01	4.633126e+00
9	1.872038e+00	67	85	3.037867e+02	2.231369e-14	1.009109e-01	4.206290e+00
10	1.791892e+00	67	85	3.078489e+02	2.213812e-14	9.094978e-02	3.118063e+00
11	1.536994e+00	67	85	3.141560e+02	2.243316e-14	7.822585e-02	2.006551e+00
12	4.387493e-01	67	85	3.323999e+02	2.858225e-14	5.541161e-02	1.015200e+00
13	3.880575e-01	67	85	3.785327e+02	3.202593e-14	5.434667e-02	9.036863e-01
14	2.979654e-01	67	85	4.665259e+02	4.320986e-14	5.297470e-02	8.772563e-01
15	2.411679e-01	67	85	5.088063e+02	4.758142e-14	5.197225e-02	8.597023e-01
16	2.118248e-01	67	85	5.511965e+02	5.084436e-14	5.092844e-02	8.415786e-01
17	1.950148e-01	67	85	5.935539e+02	5.325619e-14	4.986556e-02	8.232228e-01
18	1.840787e-01	67	85	6.357546e+02	5.467253e-14	4.879822e-02	8.048530e-01
19	1.760185e-01	67	85	6.776915e+02	5.695347e-14	4.773596e-02	7.866099e-01
20	1.694503e-01	67	85	7.192733e+02	6.273587e-14	4.668498e-02	7.690745e-01
21	1.637169e-01	67	85	7.604226e+02	6.797104e-14	4.564923e-02	7.596268e-01
22	1.584982e-01	67	85	8.010748e+02	7.280598e-14	4.463116e-02	7.502567e-01
23	1.536332e-01	68	85	8.411763e+02	7.733445e-14	4.363227e-02	7.409290e-01
24	1.490378e-01	68	85	8.806831e+02	8.161602e-14	4.265335e-02	7.316217e-01
25	1.446653e-01	68	85	9.195595e+02	8.568846e-14	4.169481e-02	7.223214e-01
26	1.404880e-01	68	85	9.577767e+02	8.957541e-14	4.075673e-02	7.130204e-01
27	1.364873e-01	68	85	9.953124e+02	9.329112e-14	3.983903e-02	7.037146e-01
28	1.326498e-01	68	86	1.032149e+03	9.684326e-14	3.894151e-02	6.944028e-01
29	1.289648e-01	68	86	1.068274e+03	1.002348e-13	3.806388e-02	6.850859e-01
30	1.254234e-01	68	86	1.103679e+03	1.034649e-13	3.720582e-02	6.757662e-01
31	1.220176e-01	68	86	1.138357e+03	1.065300e-13	3.636698e-02	6.664471e-01
32	1.187404e-01	68	86	1.172306e+03	1.094237e-13	3.554699e-02	6.571328e-01
33	1.155852e-01	68	86	1.205526e+03	1.121375e-13	3.474549e-02	6.478281e-01
34	1.125459e-01	68	86	1.238019e+03	1.146607e-13	3.396211e-02	6.385382e-01
35	1.096169e-01	68	86	1.269788e+03	1.169804e-13	3.319649e-02	6.292685e-01
36	1.067930e-01	68	86	1.300838e+03	1.190820e-13	3.244827e-02	6.200246e-01
37	1.040691e-01	68	86	1.331176e+03	1.209487e-13	3.171709e-02	6.108123e-01
38	1.014408e-01	68	86	1.360809e+03	1.225624e-13	3.100263e-02	6.016370e-01
39	9.890355e-02	68	86	1.389746e+03	1.246778e-13	3.030453e-02	5.925045e-01
40	9.645339e-02	68	86	1.417996e+03	1.272671e-13	2.962247e-02	5.834201e-01
41	9.408645e-02	68	86	1.445569e+03	1.297743e-13	2.895614e-02	5.743892e-01
42	9.179911e-02	68	86	1.472476e+03	1.322035e-13	2.830521e-02	5.654170e-01
43	8.958796e-02	68	86	1.498727e+03	1.345581e-13	2.766937e-02	5.565083e-01
44	8.744979e-02	68	86	1.524334e+03	1.368413e-13	2.704833e-02	5.476679e-01
45	8.538155e-02	68	86	1.549308e+03	1.390557e-13	2.644178e-02	5.389003e-01
46	8.338036e-02	68	86	1.573662e+03	1.412036e-13	2.584944e-02	5.302097e-01
47	8.144352e-02	68	86	1.597406e+03	1.432872e-13	2.527101e-02	5.216000e-01
48	7.956844e-02	68	86	1.620554e+03	1.453082e-13	2.470620e-02	5.130751e-01
49	7.775269e-02	68	86	1.643117e+03	1.472682e-13	2.415475e-02	5.046385e-01
50	7.599396e-02	68	86	1.665108e+03	1.491688e-13	2.361637e-02	4.962932e-01
51	7.429004e-02	68	86	1.686538e+03	1.510112e-13	2.309078e-02	4.880423e-01
52	7.263886e-02	68	86	1.707420e+03	1.527966e-13	2.257773e-02	4.798884e-01
53	7.103842e-02	68	86	1.727765e+03	1.545259e-13	2.207695e-02	4.718341e-01
54	6.948685e-02	68	86	1.747587e+03	1.562001e-13	2.158817e-02	4.638814e-01
55	6.798235e-02	68	86	1.766895e+03	1.578200e-13	2.111113e-02	4.560324e-01
56	6.652320e-02	68	86	1.785703e+03	1.593866e-13	2.064559e-02	4.482887e-01
57	6.510777e-02	68	86	1.804021e+03	1.609003e-13	2.019129e-02	4.406519e-01
58	6.373450e-02	68	86	1.821861e+03	1.623620e-13	1.974797e-02	4.331232e-01
59	6.240189e-02	68	86	1.839234e+03	1.642327e-13	1.931540e-02	4.257037e-01
60	6.110852e-02	68	86	1.856152e+03	1.680227e-13	1.889333e-02	4.183942e-01
61	5.985302e-02	68	86	1.872625e+03	1.715661e-13	1.848153e-02	4.111955e-01
62	5.863409e-02	68	86	1.888664e+03	1.748737e-13	1.807976e-02	4.041079e-01

Iter	KKT error	$N_{<10^{-6}}$	$N_{<10^{-2}}$	$\sigma_{\max}$	$\sigma_{\min}$	$\max\lvert\delta z_{1:144}\rvert$	$\max\lvert\delta z_{145:}\rvert$
63	5.745047e-02	68	86	1.904280e+03	1.779588e-13	1.768778e-02	3.971319e-01
64	5.630096e-02	68	86	1.919482e+03	1.808357e-13	1.730538e-02	3.902675e-01
65	5.518440e-02	68	86	1.934282e+03	1.835195e-13	1.693232e-02	3.835147e-01
66	5.409968e-02	68	86	1.948688e+03	1.860254e-13	1.656838e-02	3.768734e-01
67	5.304574e-02	68	86	1.962711e+03	1.883680e-13	1.621335e-02	3.703433e-01
68	5.202156e-02	68	86	1.976361e+03	1.905614e-13	1.586702e-02	3.639240e-01
69	5.102614e-02	68	86	1.989646e+03	1.926189e-13	1.552916e-02	3.576150e-01
70	5.005855e-02	68	86	2.002576e+03	1.945526e-13	1.519959e-02	3.514155e-01
71	4.911786e-02	68	86	2.015160e+03	1.963739e-13	1.487808e-02	3.453249e-01
72	4.820320e-02	68	86	2.027407e+03	1.980929e-13	1.456445e-02	3.393423e-01
73	4.731372e-02	68	86	2.039326e+03	1.997188e-13	1.425850e-02	3.334668e-01
74	4.644860e-02	68	86	2.050924e+03	2.012599e-13	1.396004e-02	3.276974e-01
75	4.560707e-02	68	86	2.062210e+03	2.027235e-13	1.366888e-02	3.220330e-01
76	4.478835e-02	68	86	2.073193e+03	2.041163e-13	1.338483e-02	3.164724e-01
77	4.399172e-02	68	86	2.083879e+03	2.054442e-13	1.310772e-02	3.110145e-01
78	4.321648e-02	68	86	2.094277e+03	2.067123e-13	1.283737e-02	3.056580e-01
79	4.246194e-02	68	86	2.104395e+03	2.079253e-13	1.257360e-02	3.004016e-01
80	4.172744e-02	68	86	2.114239e+03	2.090874e-13	1.231626e-02	2.952440e-01
81	4.101236e-02	68	86	2.123817e+03	2.102021e-13	1.206516e-02	2.901837e-01
82	4.031608e-02	68	86	2.133136e+03	2.112729e-13	1.182016e-02	2.852195e-01
83	3.963802e-02	68	86	2.142203e+03	2.123026e-13	1.158109e-02	2.803498e-01
84	3.897759e-02	68	86	2.151024e+03	2.132938e-13	1.134780e-02	2.755731e-01
85	3.833426e-02	68	86	2.159606e+03	2.142488e-13	1.112013e-02	2.708881e-01
86	3.770750e-02	68	86	2.167956e+03	2.151699e-13	1.089795e-02	2.662932e-01
87	3.709678e-02	68	86	2.176079e+03	2.160589e-13	1.068110e-02	2.617869e-01
88	3.650162e-02	68	86	2.183981e+03	2.169175e-13	1.046945e-02	2.573678e-01
89	3.592154e-02	68	86	2.191669e+03	2.177473e-13	1.026286e-02	2.530342e-01
90	3.535609e-02	68	86	2.199149e+03	2.185498e-13	1.006120e-02	2.487847e-01
91	3.480480e-02	68	86	2.206425e+03	2.193262e-13	9.864333e-03	2.446178e-01
92	3.426727e-02	68	86	2.213504e+03	2.200778e-13	9.672136e-03	2.405320e-01
93	3.374306e-02	68	86	2.220390e+03	2.208057e-13	9.484485e-03	2.365257e-01
94	3.323178e-02	68	86	2.227090e+03	2.215108e-13	9.301260e-03	2.325974e-01
95	3.273305e-02	68	86	2.233608e+03	2.221942e-13	9.122343e-03	2.287458e-01
96	3.224649e-02	68	86	2.239948e+03	2.228567e-13	8.947621e-03	2.249692e-01
97	3.177174e-02	68	86	2.246117e+03	2.234991e-13	8.776980e-03	2.212662e-01
98	3.130846e-02	68	86	2.252118e+03	2.241223e-13	8.610315e-03	2.176354e-01
99	3.085630e-02	68	86	2.257956e+03	2.247269e-13	8.447517e-03	2.140754e-01
100	3.041494e-02	68	86	2.263635e+03	2.253136e-13	8.288486e-03	2.105846e-01